

to switch on the first device. When you press any key on the remote for the second time, the first device turns off and the second device is switched on. Further pressing of the remote key

makes the devices switch on and off successively. Finally, when a key is pressed for the ninth time, the eighth device will turn off and the ninth will switch on.

Further pressing of any remote key resets IC2, which means all of its outputs, except Q0, go low. Alternatively, switch S1 can be used to reset IC2 at any time. Note that out of nine devices, only one can be switched on at a time.

Assemble the circuit on a general-

purpose PCB and enclose in a suitable cabinet. Mount the sensor in the middle of the front panel of the cabinet. Complete all the relay connections on the back side of the cabinet. ●

## LEAKAGE AND CONTINUITY TESTER

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Here is a simple, chargeable leakage and continuity tester that can be operated in two modes: normal mode and gain mode. The gain mode is especially useful for detecting small leakages in electrolytic capacitors or large resistors, which can't be detected by ordinary continuity testers without amplification of the continuity current. The normal mode is used for continuity testing.

Fig. 1 shows the circuit of leak-

age and continuity tester. The use of a rechargeable battery (3.6V, 60mAh Ni-Cd) makes the instrument so compact and lightweight that it can be easily housed inside a glue-stick tube. Amplification of current continuity is achieved by using a transistor along with small base resistor R1 (1-kilo-ohm) such that its contribution in the closed circuit is small. Resistor R2 (100-ohm) in combination with signal diode D1 limits the charging current and direction. The battery can be charged through the probes

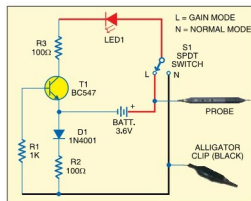


Fig. 1: Circuit of leakage and continuity tester

by connecting them to a 5V source. Use a transparent red LED (LED1) for easy detection of continuity.

Working of the circuit is simple. To check the leakage, flip S1 towards gain mode, clip the alligator clip to one terminal of the component under test, and touch the probe to the other terminal of the component. If LED1 glows, it indicates leakage.

Similarly, to check continuity, flip S1 towards normal

mode. Clip the alligator clip to ground terminal of the circuit and touch the probe to the terminal that you want to check for continuity/shorting. If LED1 glows with high intensity, it indicates continuity.

The proposed arrangement for this leakage and continuity tester is shown in Fig. 2. First, in the glue-stick tube, make the necessary holes and a rectangular slot for the positive probe tip, LED1 and single-pole double-throw switch (SPDT) S1, respectively. Complete the entire wiring, except for the alligator clip and the switch, as shown in the schematic diagram. Pull the wires for the alligator clip and the switch outside the glue-stick tube and then finish their soldering. Fasten the positive gel-pen probe tip

